WASP-36b

R-0992

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-36_180226` · created 2026-07-29T23:36:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458175.6704 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.063 +/- 0.035
Transit depth (Rp/Rs)^2	0.39 +/- 0.44 [%]
Orbital Inclination (inc)	87.97 +/- 3.04
Airmass coefficient 1 (a1)	0.971 +/- 0.015
Airmass coefficient 2 (a2)	0.0197 +/- 0.0067
Scatter in the residuals of the lightcurve fit is	1.5 %
Best Comparison Star	#3 - [548, 26]
Optimal Method	PSF photometry
Transit Duration (day)	0.0837 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.063 ± 0.035 vs 0.1368 (deviation 2.11σ)
Duration fit vs published	0.0837 d vs 0.0758 d (deviation 0.81σ)
Mid-time O–C	-3.2 min
Depth SNR	1.8
Residual scatter	1.5 %

Light curve

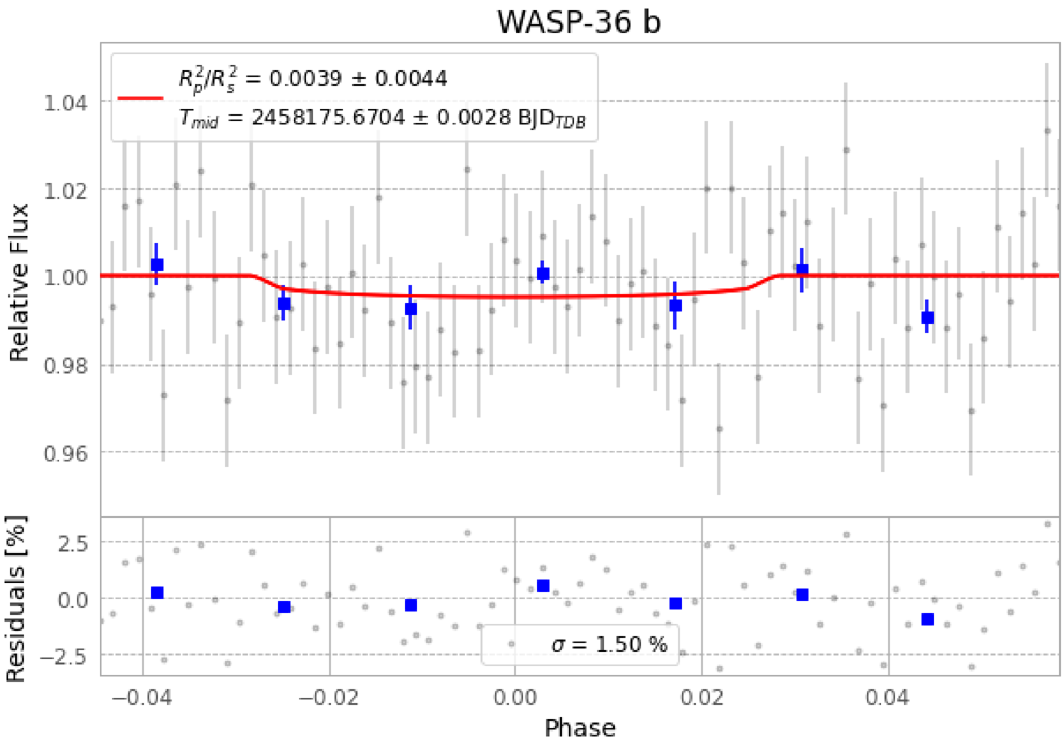

Figure 1 — FinalLightCurve_WASP-36 b_2018-02-25.png (SHA-256 b353a5bf27ec6c87…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [395, 220], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 d0129dd525b75887…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

77 FITS frames (51 MB), `WASP-36180226021812.FITS` → `WASP-36180226060612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-180226.log (2efff3020663…), FinalLightCurve_WASP-36 b_2018-02-25.png (b353a5bf27ec…), FinalLightCurve_WASP-36 b_2018-02-25.csv (af214a40a1b2…)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)